CNNs vs. Vision Transformers under Geographic Domain Shift: From EuroSAT to Semi-Arid Aguascalientes, Mexico

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***Abstract*—Land use in Aguascalientes, a semi-arid state in central Mexico, is tied to its main water problem, the overexploitation of groundwater. Yet no labeled Sentinel-2 patches of the state existed, and available land-use models were trained on European landscapes. We measured how much such a model loses there and how much light fine-tuning wins back. We compared two convolutional neural networks (CNNs), ResNet-50 and EfficientNet-B0, with a Vision Transformer, ViT-S/16, in eight runs under one shared protocol, with RGB or 13-band input and ImageNet or self-supervised SSL4EO-S12 weights. The models were trained on EuroSAT, then fine-tuned and tested on $1{,}102$ hand-labeled Aguascalientes patches (561 for fine-tuning, 541 for testing), split by spatial blocks. Every run reached a EuroSAT macro-F1 of at least 0.983. Without local training, no Aguascalientes macro-F1 went above 0.354, and fine-tuning the last block recovered at most 0.268 of the drop. In Aguascalientes, 13 bands helped only ResNet-50. With 13 bands, SSL4EO-S12 weights scored higher than ImageNet weights after fine-tuning for both ResNet-50 and ViT-S, but a different normalization prevents crediting the weights alone. Among the ImageNet runs, ViT-S/RGB had the highest observed macro-F1 after fine-tuning, 0.441; with one seed, and a last block that holds a different share of each model, this is not a firm ranking of CNNs and transformers. A selection rule fixed in advance chose ResNet-50 with 13 bands and SSL4EO-S12 weights, which reached 0.525 after fine-tuning. In other words, a near-perfect EuroSAT score does not guarantee a useful model in another region.**

***Index Terms*—Convolutional neural networks, domain shift, land-use classification, remote sensing, Sentinel-2, vision transformers**

I. Introduction

Our goal was to measure how much a land-use classifier trained on European Sentinel-2 images loses when it is used in Aguascalientes, a semi-arid state in central Mexico, and how much of that loss a small set of local labels can win back. We compared two convolutional neural networks (CNNs), ResNet-50 and EfficientNet-B0, with a Vision Transformer, ViT-S/16. All three were trained on EuroSAT and then tested on patches of Aguascalientes that we labeled by hand.

Land use matters in Aguascalientes because of water. The state water plan names the overexploitation of groundwater, the main source of supply, as the central problem of the state [1]. The same plan states that the demand for water comes mainly from agricultural and urban public use. Both uses can be seen from a satellite: farmland and the city are land-use classes that a Sentinel-2 image can show. Monitoring land use is therefore one way to follow the pressure on water, although it does not measure the water itself.

Monitoring land use in Aguascalientes faced a practical problem. There were no labeled Sentinel-2 patches of Aguascalientes, and the land-use models available had been trained on European landscapes such as those of EuroSAT. The long-term goal of the project is a monthly land-use monitor of the state, which is the second stage of this work. That monitor needs a model that works in Aguascalientes, so this paper chooses one and measures how well it transfers.

The difficulty has a name: geographic domain shift. A model learns how each class looks in the region where it was trained, and a new region can show the same class with other colors and textures. A model that learned Forest as a dark, closed canopy and Pasture as green grass has little reason to recognize their versions in semi-arid Aguascalientes, imaged in April, in the dry season [1]. The size of this loss can only be measured fairly on a test set that is spatially separated from the labels used to adapt the model.

We organized the study around five research questions. Which of ResNet-50, EfficientNet-B0 and ViT-S/16 performs best on EuroSAT, and at what computational cost (RQ-a)? How much does each model degrade on Aguascalientes without any local training, that is, zero-shot (RQ-b)? How much of that degradation does light fine-tuning recover (RQ-c)? Does 13-band input outperform RGB (RQ-d)? And do self-supervised SSL4EO-S12 weights outperform ImageNet weights (RQ-e)?

The short answer is that the domain shift outweighed the differences between the models. On EuroSAT, the eight runs reached a macro-F1, the average of the per-class F1 scores, between 0.983 and 0.993. In Aguascalientes, without local training, they lost between 0.639 and 0.788 in macro-F1, and fine-tuning the last block recovered at most 0.268 of that drop (Eq. 3). The highest observed macro-F1 after fine-tuning, 0.525, came from ResNet-50 with 13 bands and SSL4EO-S12 weights, the model we selected.

Our contributions are the following:

- A comparison of ResNet-50, EfficientNet-B0 and ViT-S/16 under one shared training protocol, including RGB

versus 13-band input and ImageNet versus SSL4EO-S12 weights.
- A measurement of the drop from EuroSAT to Aguascalientes and of its partial recovery with two light fine-tuning modes, on 1,102 hand-labeled patches (561 for fine-tuning and 541 for testing) split by blocks of 8×8 cells (5.12 km).
- Open code, labels and checkpoints, so that the experiments can be repeated and the models reused.

The rest of the paper is organized as follows. Section II reviews related work. Section III describes the two datasets and the spatial split, and Section IV the models, the shared protocol and the metrics. Section V reports the results, Section VI discusses their likely causes and limits, and Section VII concludes.

II. Related Work

Convolutional neural networks (CNNs) are the usual starting point for land-use classification of Sentinel-2 patches. EuroSAT [2], the dataset we trained on, was introduced together with a benchmark of deep CNNs over its 13 spectral bands and 10 classes, and those CNNs reached a high overall accuracy. BigEarthNet [3] is a much larger Sentinel-2 archive in which each patch has several land-cover labels. Its authors found that a shallow CNN trained on it gave much higher accuracy than a state-of-the-art CNN pre-trained on ImageNet, a result that suggests that features learned on natural photographs do not fully transfer to Sentinel-2 images. Because both benchmarks were introduced with CNN baselines, CNNs are the natural reference when a new model is tested on Sentinel-2 patches. We used two CNN families. ResNet [4] learns residual functions, which makes very deep networks easier to optimize. EfficientNet [5] scales depth, width and input resolution together with one compound coefficient, and its authors report better accuracy and efficiency than previous CNNs.

Transformers reached image classification later. The Vision Transformer (ViT) [6] applies a pure transformer to a sequence of image patches, without convolutions. When pre-trained on large amounts of data, it obtained excellent results compared with state-of-the-art CNNs while needing fewer computational resources to train. Bazi et al. [7] applied ViT to remote-sensing scene classification. In their model, multihead attention derives long-range relations between pixels, and on four remote-sensing datasets the model showed promising results compared with state-of-the-art methods. SatMAE [8] went one step further and pre-trained transformers for temporal or multispectral satellite images with a masked autoencoder. Its authors found it beneficial to encode the spectral bands as groups with their own spectral position encodings, and they report gains in transfer learning, including land-cover classification. This suggests that a transformer can use extra spectral bands when its pre-training is designed for them, which is relevant to our 13-band runs.

Self-supervised pre-training addresses a different gap: labels are scarce, while unlabeled satellite images are plentiful. A review by Wang et al. [9] introduces self-supervised learning for remote sensing and notes that most of its potential in Earth observation remains locked. Seasonal Contrast (SeCo) [10] starts from the same observation that motivates our study: models pre-trained on ImageNet may not generalize to remote sensing because of the domain gap. SeCo gathers unlabeled images of many locations at different times and uses invariance to time and position to learn transferable representations; its models outperformed their ImageNet counterparts on several downstream tasks. SSL4EO-S12 [11] provides a large, global and multiseasonal corpus of Sentinel-1 and Sentinel-2 images, with pre-trained models for methods such as MoCo and DINO. We used its ResNet-50 and ViT-S/16 weights as the self-supervised alternative to ImageNet.

The last line of work concerns what happens when the test region differs from the training region. Tuia et al. [12] note that when training samples come from a different spatial region than the one being mapped, spectral shifts between the two distributions are likely to make the model fail. They attribute such shifts to differences in acquisition and atmospheric conditions or to changes in the observed objects, and they review domain adaptation methods designed to be robust to such shifts. WILDS [13] collects real distribution shifts, including shifts across time and location in satellite imaging, and shows that standard training gives much lower performance out of distribution than in distribution. Measuring such a shift also requires an honest test set. Roberts et al. [14] show that random cross-validation on spatially structured data can seriously underestimate the prediction error, and they recommend block cross-validation, where the data are split by blocks instead of at random. These studies leave a practical question open for our setting: how two CNNs and a ViT, trained under one shared protocol, behave when they move from European Sentinel-2 patches to a semi-arid region of Mexico, measured on hand-labeled patches split by spatial blocks.

III. Data

We used two datasets with the same patch format: EuroSAT to train the models in Europe, and a new set of labeled patches from Aguascalientes to fine-tune and test them. We kept the format identical so that the main thing that changes between the two is the landscape.

A. EuroSAT

We used EuroSAT [2], a dataset of Sentinel-2 [15] images in 13 spectral bands and 10 land-use classes. Its multispectral version, EuroSAT MS, holds L1C patches of 13×64×64 pixels at 10 m. We kept the official TorchGeo [16] split of 16,200 training, 5,400 validation and 5,400 test patches, and the TorchGeo band order (B01 to B12, with B8A last). The RGB input is B04, B03 and B02 taken from the same 13-band files. We chose this so that the RGB and 13-band runs see the same patches and the same pixel values; only the number of bands changes. With a separately processed RGB version, part of any

gap between the two inputs could come from that processing instead of from the bands.

B. Aguascalientes composite and grid

We built one Sentinel-2 composite of the state for April 2024 in Google Earth Engine [17]. We used the L1C harmonized collection (`COPERNICUS/S2_HARMONIZED`, digital numbers equal to reflectance $\times$ 10,000) because EuroSAT is also L1C. With surface reflectance (L2A), the models would face a change of processing level on top of the change of region. We chose April because it falls in the dry season, and March and April are the months with the least rain of the year [1], so one month gives enough clear images. Clouds were masked with QA60 bits 10 (opaque) and 11 (cirrus), and we took the per-pixel median of the month. The median reduces the clouds and shadows that the mask misses, because they are outliers in the time series of each pixel. All bands were resampled to 10 m and clipped to the state boundary of FAO GAUL 2025 level 1.

We cut the composite with an absolute 640 m grid in EPSG:32613 and extracted one $13\times64\times64$ patch per cell, in the EuroSAT band order. The grid size follows from EuroSAT: 64 pixels of 10 m cover 640 m, so each cell is exactly one EuroSAT-sized patch. The grid origin lies on multiples of 640 m, so no pixel is split between two cells.

C. Labels and spatial split

A single annotator labeled the cells with the 10 EuroSAT classes on an RGB preview of the composite. The April 2024 preview was the main evidence; the Esri World Imagery basemap was used only to recognize objects such as roofs, tree rows and roads, because it is recent imagery, often from another season. A cell got a class only if at least 70 % of it was that class; Highway and River were labeled by their dominant element. A labeling guide fixed tie-breakers for semi-arid confusions: xeric scrub is HerbaceousVegetation, managed grazing grass is Pasture, continuous canopy is Forest, and trees or vines in rows are PermanentCrop. We refined the guide after the first 547 cells and re-labeled those cells with the final version (details in the data card). Cells whose cover has no EuroSAT class (greenhouses, quarries, bare rock or soil) or that raised any doubt were skipped and never used. This reduces label noise in the test set, but it also means the test set does not measure errors on those covers.

Neighboring cells are almost identical, so we split by space and not by cell [14]. A random cell split would put near-duplicates in the fine-tune and test sets and inflate the scores. Instead, we grouped the cells into blocks of 8×8 cells (5.12 km), and each block went whole to one split. With seed 0, we shuffled the block list 20,000 times, cut it half/half, and kept the partition with the smallest maximum deviation of any class from 50 % in test. That deviation was 0.0294, so every class is split almost evenly; each split holds 119 blocks. The split has no buffer zone, which leaves a residual risk: 35 of the 541 test patches border a fine-tune cell across a block edge, so the fine-tuned scores may be slightly optimistic.

TABLE I
LABELED AGUASCALIENTES PATCHES PER CLASS AND SPLIT. †FEWER THAN 10 TEST PATCHES: LEFT OUT OF THE AGUASCALIENTES MACRO-F1 AND REPORTED SEPARATELY.

Class	Fine-tune	Test	Total
AnnualCrop	109	104	213
Forest	45	40	85
HerbaceousVegetation	245	236	481
Highway	33	31	64
Industrial	24	24	48
Pasture	24	26	50
PermanentCrop	24	26	50
Residential	38	34	72
River†	7	7	14
SeaLake	12	13	25
Total labeled	561	541	1,102
Skipped (not used)	85	83	168

Table I lists the labeled patches per class and split. It shows a strong imbalance: HerbaceousVegetation has 236 test patches and SeaLake only 13. River has 7, fewer than 10, so we left it out of the Aguascalientes macro-F1 and report it separately. The last row counts the 168 skipped cells, which appear in no experiment. Fig. 1 shows the blocks on the composite and compares patches of the same class from both regions. The gap is visible before any model is trained.

IV. METHODOLOGY

We trained three pre-trained models on EuroSAT under one shared protocol, tested them on Aguascalientes without local training, and then fine-tuned them on the Aguascalientes fine-tune split. This section describes the models, the protocol, the experiments and the metrics.

A. Models and initialization

We compared ResNet-50 [4], a CNN whose layers learn residual functions; EfficientNet-B0 [5], the baseline of a CNN family scaled in depth, width and resolution; and ViT-S/16 [6], a pure transformer applied to a sequence of 16×16-pixel image patches. This gives two CNN families and one transformer that can all be trained on one desktop machine with the same settings. We chose the small ViT because its size is close to that of ResNet-50: with RGB input it has 21.7 M parameters against 23.5 M, while EfficientNet-B0 has 4.0 M (Table VI). A CNN versus ViT gap is therefore not just a gap in model size.

All three models started from ImageNet [18] weights (timm). For ResNet-50 and ViT-S/16 we also used SSL4EO-S12 weights [11], learned without labels on the Sentinel-2 L1C images of the SSL4EO-S12 corpus with self-supervised methods that include MoCo and DINO. We used the MoCo weights for ResNet-50 and the DINO weights for ViT-S/16. These weights exist only for the 13 L1C bands, and they expect TorchGeo's normalization, $x/10{,}000$. EfficientNet-B0 has no SSL4EO-S12 weights, so it has only ImageNet runs. The SSL4EO-S12 band order puts B8A right after B08, while our data keep B8A last. We therefore permuted the input channels

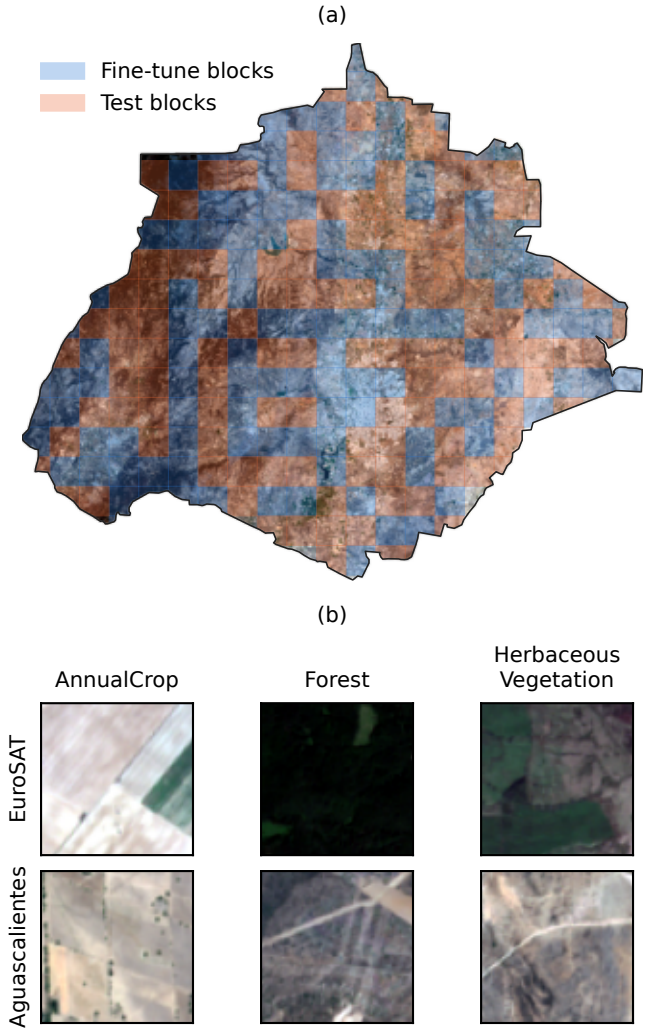


Fig. 1. (a) April 2024 RGB composite of Aguascalientes with the 8×8-cell blocks colored by split (blue: fine-tune, orange: test) and the state outline. (b) One EuroSAT test patch (top row) and one Aguascalientes test patch (bottom row) for AnnualCrop, Forest and HerbaceousVegetation; they are the patches of the Grad-CAM figures. RGB stretch 2–98 % per dataset; Aguascalientes uses the global limits of the composite, as the labeling previews did. Note how differently the same class looks in Europe and in semi-arid Aguascalientes.

of the first layer to EuroSAT order; without this step, B8A and B09 to B12 would each enter through a filter learned for a different band.

ImageNet models expect three input channels, so the 13-band runs needed a new first layer (the first convolution, or the patch embedding of the ViT). We copied the pre-trained RGB filters to B04, B03 and B02 and set every other channel to the mean of the three. In this way, the RGB filters stay exactly as pre-trained, and the new channels start from a neutral filter instead of random noise that would disturb the pre-trained features in the first epochs.

B. Shared training protocol

The three models share everything except the architecture, as Table II shows. We never changed a hyperparameter for one model only, so that differences come from the architecture and not from tuning. For example, if only the ViT had received a tuned learning rate, a ViT advantage could be the result of that tuning. Patches were resized from 64 to 224 pixels because this is the input size of the pre-trained weights. ImageNet runs used the per-band mean and standard deviation of EuroSAT train, also on Aguascalientes, so that no statistic of the test region enters the pipeline. Every run used seed 0 and was trained once, so we have no variance estimate across seeds.

TABLE II
CONFIGURATION SHARED BY EVERY RUN, FROM THE BASE CONFIGURATION FILE. ONLY THE ARCHITECTURE, THE INPUT BANDS AND THE INITIAL WEIGHTS CHANGE BETWEEN RUNS.

Setting	Value
Input	64×64 resized to 224×224 (bilinear)
Normalization	ImageNet: per-band mean and std of EuroSAT train; SSL4EO-S12: $x/10{,}000$ (TorchGeo)
Augmentation	horizontal flip, vertical flip, 90° rotations
Optimizer	AdamW, lr 1×10^{-4}, weight decay 0.05
Schedule	1 warmup epoch + cosine
Batch size	64 (2×32, accumulation)
Epochs (EuroSAT)	max 15, patience 3 on val macro-F1
Loss	EuroSAT: cross-entropy; Aguascalientes: class-weighted cross-entropy
Fine-tuning	modes `head`, `last_block`; 10 fixed epochs, last epoch kept
Seed	0

We ran every experiment on the same Mac mini with an Apple M2 Pro, using the PyTorch MPS backend in fp32. Because of memory, the effective batch of 64 was built from two micro-batches of 32 with gradient accumulation; BatchNorm therefore saw 32 samples at a time.

C. Experiments

The study has four experiments. In E1 we trained the three models on EuroSAT with ImageNet weights, each with RGB and with 13 bands (6 runs). In E2 we trained ResNet-50 and ViT-S/16 with SSL4EO-S12 weights and 13 bands (2 runs). In E3 we evaluated the 8 resulting checkpoints on the Aguascalientes test split without any local training, that is, zero-shot. In E4 we fine-tuned the 8 checkpoints on the fine-tune split in two modes (16 runs). Because SSL4EO-S12 runs also use a different normalization, a gap between SSL4EO-S12 and ImageNet runs cannot be credited to the weights alone.

The two fine-tuning modes differ in how much of the network they train. In `head` mode, the backbone was frozen and only the classifier was trained. In `last_block` mode, the last stage or block was trained together with the classifier. Frozen parts ran in evaluation mode, so BatchNorm kept its EuroSAT statistics and dropout was off. To keep rare classes from being ignored, the cross-entropy loss used the class weights

$$w_c = \frac{n}{K\, n_c}, \tag{1}$$

where n is the number of fine-tune patches, K is the number of classes and n_c is the number of fine-tune patches of class c. The weights came from the fine-tune split only. Fine-tuning ran for a fixed 10 epochs and we kept the last epoch. We chose

fixed epochs because there is no validation split in Aguascalientes: early stopping on the test split would leak test information into the model. One confound remains: `last_block` trains a different share of the parameters of each architecture, so a CNN versus ViT gap after `last_block` cannot be credited to the architecture alone.

D. Metrics

Our main metric is macro-F1, the average of the per-class F1 scores. With it, a rare class counts as much as a common one; with accuracy, a model could score well on Aguascalientes by predicting HerbaceousVegetation for most patches. The Aguascalientes macro-F1 averages the 9 classes with at least 10 test patches, so River is reported separately. We also report the macro-F1 over all 10 classes and the accuracy. For cost, we recorded parameters, training time, peak memory, inference throughput (patches/s) and the epochs needed to reach 95 % of the best validation macro-F1.

To measure the domain shift, we defined the drop and the recovery as

$$\mathrm{Drop} = \mathrm{F1}_{\mathrm{EuroSAT}} - \mathrm{F1}_{\mathrm{Ags}}^{\text{zero-shot}}, \tag{2}$$

$$\mathrm{Recovery} = \frac{\mathrm{F1}_{\mathrm{Ags}}^{\text{fine-tuned}} - \mathrm{F1}_{\mathrm{Ags}}^{\text{zero-shot}}}{\mathrm{Drop}}, \tag{3}$$

where $\mathrm{F1}_{\mathrm{EuroSAT}}$ is the macro-F1 over the same 9 classes, computed from the per-class F1 on the EuroSAT test set. In other words, the drop is what a model loses when it moves to Aguascalientes, and the recovery is the fraction of that loss that fine-tuning wins back. Recovery is undefined when $\mathrm{Drop} \leq 0$.

We fixed the selection rule before seeing any result. The selected model is the one with the highest Aguascalientes macro-F1 after `last_block` fine-tuning. If two models are within 0.01 macro-F1, the one with the higher patches/s on the EuroSAT test set wins. Freezing the rule first prevents choosing the criterion that favors a preferred model after the fact.

E. Grad-CAM

To see which parts of a patch drive a prediction, we used Grad-CAM [19]. We computed it for the predicted class on the 224×224 input, resized the map to the 64×64 patch and scaled it to $[0, 1]$. The target layer was the last block of `layer4` for ResNet-50 and `conv_head` for EfficientNet-B0, the last convolutional features of each CNN. For ViT-S/16 we used the first LayerNorm (`norm1`) of the last block. There, we dropped the CLS token and reshaped the 196 patch tokens into a 14×14 map, so the gradients have a spatial layout. We chose this layer because it is the last point where the patch tokens still affect the prediction: after the last attention layer, only the CLS token reaches the classifier, so the gradients on the patch tokens would be zero.

V. Experimental Results

On EuroSAT the eight runs were almost tied; in Aguascalientes all of them lost most of their macro-F1, and fine-tuning won back only part of it. We report the results in the order of the experiments: EuroSAT, Aguascalientes without and with fine-tuning, the input and weight variants, the cost, the selected model and the Grad-CAM maps. Every configuration was trained once with seed 0, so we report the highest observed values and make no claim that one model is better than another.

TABLE III
EuroSAT test set (10 classes): overall accuracy and macro-F1 of the eight runs. Best value per column in bold.

Run	Accuracy	Macro-F1
ResNet-50 / RGB / ImageNet	0.984	0.983
ResNet-50 / MS13 / ImageNet	0.985	0.985
ResNet-50 / MS13 / SSL4EO	**0.993**	**0.993**
EfficientNet-B0 / RGB / ImageNet	0.984	0.984
EfficientNet-B0 / MS13 / ImageNet	0.985	0.985
ViT-S / RGB / ImageNet	0.991	0.991
ViT-S / MS13 / ImageNet	0.990	0.990
ViT-S / MS13 / SSL4EO	0.987	0.987

A. EuroSAT

All eight runs reached almost the same score on the EuroSAT test set. Table III shows that macro-F1 ranged from 0.983 to 0.993, and accuracy followed the same pattern. ResNet-50/MS13/SSL4EO had the highest observed value in both columns. This means that EuroSAT alone does not separate the models.

The training curves tell the same story. Fig. 2 shows the validation macro-F1 per epoch: every run reached a high value within the first few epochs and changed little afterwards. The curves differ mainly in the first epoch, where the two ResNet-50 runs with ImageNet weights started lowest. With 13 bands, the SSL4EO-S12 run of each model started higher than the ImageNet run of the same model, although ViT-S/RGB/ImageNet started slightly above ViT-S/MS13/SSL4EO. The epochs needed to reach 95 % of the best validation macro-F1 did not separate the models either: every run needed at most 2 epochs.

B. Aguascalientes without fine-tuning

The models lost most of their EuroSAT performance when we tested them on Aguascalientes without local training. The zero-shot column of Table IV shows macro-F1 between 0.196 and 0.354 over the nine classes. The drop, the EuroSAT F1 minus the zero-shot F1, ranged from 0.639 to 0.788. In other words, most of what the models learned in Europe did not carry over to semi-arid Aguascalientes.

Fig. 3(a) shows where the loss happened. HerbaceousVegetation had the highest per-class F1 in every run, while Forest and Pasture were at or near zero in every run. River appears hatched because it has only seven test patches and is reported separately. The pattern is the same for the three architectures, both inputs and both kinds of weights.

The confusion matrix shows the same collapse from the side of the predictions. Fig. 4(a) gives the zero-shot confusion matrix of the model we selected later (Section V-F). Most errors fell in the HerbaceousVegetation and PermanentCrop

TABLE IV

MACRO-F1 ON THE AGUASCALIENTES TEST SPLIT. THE CLASS SET IS THE 9 CLASSES WITH AT LEAST 10 TEST PATCHES (1 CLASS EXCLUDED AND REPORTED SEPARATELY). EUROSAT F1 IS THE MACRO-F1 OF THE EUROSAT TEST RESULT OVER THE SAME CLASSES. DROP = EUROSAT F1 MINUS ZERO-SHOT; RECOVERY = (FINE-TUNED MINUS ZERO-SHOT) / DROP, SHOWN AS – WHEN THE DROP IS NOT POSITIVE. BEST F1 PER COLUMN IN BOLD.

Run	EuroSAT F1	Zero-shot	Head	Last block	Drop	Rec. head	Rec. last block
ResNet-50 / RGB / ImageNet	0.983	0.259	0.308	0.278	0.724	0.068	0.026
ResNet-50 / MS13 / ImageNet	0.984	0.271	0.339	0.371	0.713	0.095	0.140
ResNet-50 / MS13 / SSL4EO	**0.993**	**0.354**	0.372	**0.525**	0.639	0.029	0.268
EfficientNet-B0 / RGB / ImageNet	0.984	0.271	0.275	0.400	0.713	0.006	0.181
EfficientNet-B0 / MS13 / ImageNet	0.984	0.196	0.217	0.361	0.788	0.027	0.209
ViT-S / RGB / ImageNet	0.991	0.326	**0.380**	0.441	0.665	0.081	0.173
ViT-S / MS13 / ImageNet	0.989	0.287	0.340	0.361	0.703	0.076	0.106
ViT-S / MS13 / SSL4EO	0.987	0.287	0.305	0.428	0.699	0.025	0.201

columns; for example, every Forest test patch was predicted as HerbaceousVegetation. This means that, without local training, the predictions concentrated in two classes, one of them the largest class of the test set.

C. Fine-tuning on Aguascalientes

Fine-tuning won back only part of the drop, and how much depended on the mode. The Head and Last block columns of Table IV give the macro-F1 after each mode, and Fig. 5 shows the drop (a) and the recovery (b) of every run. With `head`, the recovery was at most 0.095, so training only the classifier barely changed the scores. With `last_block`, the recovery ranged from 0.026 to 0.268. This means that the largest part of the drop remained after fine-tuning in every run.

One run did not follow the general order of the two modes. In every run except ResNet-50/RGB/ImageNet, `last_block` gave a higher macro-F1 than `head`. For ResNet-50/RGB/ImageNet, macro-F1 after `last_block` (0.278) was below that after `head` (0.308). Its Aguascalientes accuracy also fell, from 0.542 zero-shot to 0.287 after `last_block`.

The fall in accuracy was not limited to that run. Table V gives the accuracy of every run, that is, the share of test patches classified correctly, zero-shot and after each mode. In 7 of the 8 runs, `last_block` lowered the Aguascalientes accuracy below its zero-shot value while it raised macro-F1. This trade-off is consistent with the class-weighted loss of Eq. (1), which shifts predictions away from the dominant HerbaceousVegetation class. Fig. 3(b) shows both sides of it: Forest and Pasture are no longer at zero in any run, and HerbaceousVegetation F1 is lower than zero-shot in every run.

D. Input bands and pre-trained weights

Whether 13 bands helped depended on the model. Comparing the RGB and 13-band runs with ImageNet weights in Table IV, only ResNet-50 scored higher with 13 bands, both zero-shot and after `last_block`. EfficientNet-B0 and ViT-S scored lower with 13 bands in both columns. In other words, our results do not support a general rule that more bands help under this domain shift.

With 13 bands, the SSL4EO-S12 runs reached a higher macro-F1 after `last_block` than the ImageNet runs, for both ResNet-50 and ViT-S. Zero-shot, ResNet-50 was also higher with SSL4EO-S12, while ViT-S gave the same value at three decimals with both kinds of weights. On EuroSAT, ViT-S/MS13/SSL4EO was slightly lower than its ImageNet counterpart (Table III). Because the SSL4EO-S12 runs also used a different normalization, $x/10{,}000$, these gaps cannot be credited to the weights alone. In the same way, `last_block` trains a different share of the parameters of each architecture, so we leave the CNN versus ViT comparison after fine-tuning to Section VI.

TABLE V

ACCURACY ON THE AGUASCALIENTES TEST SPLIT: THE SHARE OF THE 541 TEST PATCHES (ALL 10 CLASSES) CLASSIFIED CORRECTLY, ZERO-SHOT AND AFTER EACH FINE-TUNING MODE. BEST VALUE PER COLUMN IN BOLD.

Run	Zero-shot	Head	Last block
ResNet-50 / RGB / ImageNet	0.542	0.562	0.287
ResNet-50 / MS13 / ImageNet	0.484	0.484	0.440
ResNet-50 / MS13 / SSL4EO	**0.593**	**0.608**	**0.579**
EfficientNet-B0 / RGB / ImageNet	0.521	0.525	0.484
EfficientNet-B0 / MS13 / ImageNet	0.353	0.357	0.405
ViT-S / RGB / ImageNet	0.553	0.586	0.499
ViT-S / MS13 / ImageNet	0.455	0.468	0.370
ViT-S / MS13 / SSL4EO	0.536	0.545	0.499

E. Computational cost

Table VI lists the cost of the eight EuroSAT runs. Peak memory is the memory reserved by the MPS driver, not the memory that the weights and activations of the model need on their own. ViT-S reserved the most: at least 5.18 GiB, against at most 4.33 GiB for the CNNs. Inference throughput ranged from 183 to 236 patches/s, and each 13-band run had a lower throughput than the RGB run of the same model.

Total training time is a weak basis for comparison, because early stopping ended the runs at different epochs (Fig. 2). Time per epoch is the comparable column, and it was similar across the three models. EfficientNet-B0 has far fewer parameters than ResNet-50 and ViT-S, yet its time per epoch was close to theirs. This means that, on our hardware, the number of parameters did not predict the training time.

F. Selected model

Applying the selection rule fixed in Section IV, we selected ResNet-50/MS13/SSL4EO with `last_block` fine-tuning. Its

TABLE VI
TRAINING COST OF THE EIGHT EUROSAT RUNS ON AN APPLE M2 PRO (FP32). PEAK MEMORY IS THE MEMORY RESERVED BY THE MPS DRIVER. PATCHES/S IS INFERENCE THROUGHPUT ON THE EUROSAT TEST SET.

Run	Params (M)	Train time (min)	Time/epoch (s)	Peak mem. (GiB)	Patches/s
ResNet-50 / RGB / ImageNet	23.5	58.3	233	4.31	208
ResNet-50 / MS13 / ImageNet	23.6	42.0	252	4.31	184
ResNet-50 / MS13 / SSL4EO	23.6	62.9	252	4.31	183
EfficientNet-B0 / RGB / ImageNet	4.0	45.5	227	4.33	233
EfficientNet-B0 / MS13 / ImageNet	4.0	57.8	231	4.33	214
ViT-S / RGB / ImageNet	21.7	54.6	218	5.18	236
ViT-S / MS13 / ImageNet	22.7	55.3	237	5.25	199
ViT-S / MS13 / SSL4EO	22.7	59.3	237	5.25	199

Aguascalientes macro-F1 was 0.525, 0.084 above the runner-up, which is wider than the tie-break margin, so the throughput rule was not needed. Since only ResNet-50 and ViT-S have SSL4EO-S12 runs, we also applied the rule to the six ImageNet runs as a secondary result, where the three models compete on equal weights. There, ViT-S/RGB/ImageNet came first with 0.441, 0.040 above the next run, again without a tie-break.

Fig. 4(b) shows the selected model after `last_block`. Compared with panel (a), the predictions now spread along the diagonal instead of piling up in the HerbaceousVegetation column. The largest off-diagonal cell is HerbaceousVegetation predicted as Forest. In other words, the main error changed direction: from most patches being called HerbaceousVegetation to part of HerbaceousVegetation being called Forest.

G. Grad-CAM

Fig. 6 shows the Grad-CAM maps of the RGB ImageNet run of each model, without fine-tuning, on EuroSAT (a) and Aguascalientes (b). We drew one test patch per class at random with seed 0, for six classes, so the patches were not chosen to make a point; the list and the rule are stored with the figures. Under each map is the predicted class, marked as correct or wrong. On EuroSAT all predictions but one are correct, while on Aguascalientes most are wrong, and most of the wrong ones are HerbaceousVegetation.

The two families of models produce maps of a different shape. CNN maps are smooth and concentrated on one compact region of the patch, sometimes two. ViT-S maps are made of many small spots spread over the whole patch. In Aguascalientes, many maps fall away from the object that defines the class. The Industrial cell of Fig. 6(b) is a solar park, seen as a dark rectangle of panels at the top left. The maximum of the ResNet-50 map lies on the edge of the panels. The maximum of EfficientNet-B0 falls in the lower-left corner, and the spots of ViT-S spread over the bare ground. None of the three models predicted Industrial for this patch.

Fig. 7 compares the selected model before and after `last_block` on the Aguascalientes patches of Fig. 6(b). The HerbaceousVegetation and SeaLake rows are replaced by two HerbaceousVegetation patches drawn at random with seed 0 among those that the fine-tuned model predicts as Forest; that is, they were chosen to show the main error of Fig. 4(b). Both were correct zero-shot and became Forest after `last_block`. The AnnualCrop patch changed from a wrong prediction to a correct one, the Forest and Industrial patches stayed wrong, and the Residential maps are almost the same before and after. We leave the interpretation of all these maps to Section VI.

VI. DISCUSSION

Section V described what happened; this section discusses why. Every configuration was trained once, and no experiment was designed to isolate a cause, so each explanation below is a hypothesis that fits the evidence, not a proven mechanism. We follow the research questions: the drop (RQ-b), fine-tuning (RQ-c), the architectures (RQ-a), and the input bands and weights (RQ-d, RQ-e). We then discuss the Grad-CAM maps, the cost and the limitations.

A. Why the drop was so large (RQ-b)

We could observe two sources of the drop: the pixels changed, and the classes looked different. To check the pixels, we compared the median of each band of the April composite with the EuroSAT training statistics as a z-score, $z = (\tilde{x}_{\mathrm{Ags}} - \mu_{\mathrm{EuroSAT}})/\sigma_{\mathrm{EuroSAT}}$. The visible bands (B02–B04) lay 0.95 to 1.43 standard deviations above the EuroSAT mean and the SWIR bands (B11, B12) 1.50 to 1.91 above it, while the near-infrared bands (B08, B8A) stayed within 0.20. This pattern is consistent with a landscape of bare soil and dry vegetation, which reflect more visible and SWIR light than the green fields and forests of EuroSAT. Normalization does not remove this shift, because it applies the same EuroSAT statistics (or, for SSL4EO-S12, the same fixed scale) to both datasets. In other words, a typical Aguascalientes pixel looked to the network like an unusually bright EuroSAT pixel.

B10 was the only band that this screening flagged, with $z = 6.68$. B10 is the cirrus band: it sits in a strong water-vapor absorption band and sees little of the ground, so one possible reason is the atmosphere, a drier air column over the high plateau or thin cirrus that the cloud mask missed, rather than a different land cover; we did not test it. Since the screening uses one median per band for the whole state, it says nothing about individual classes.

The second source is that the same class does not look the same. Fig. 1(b) shows it: the EuroSAT Forest patch is a dark, closed canopy, while the Aguascalientes one is a gray,

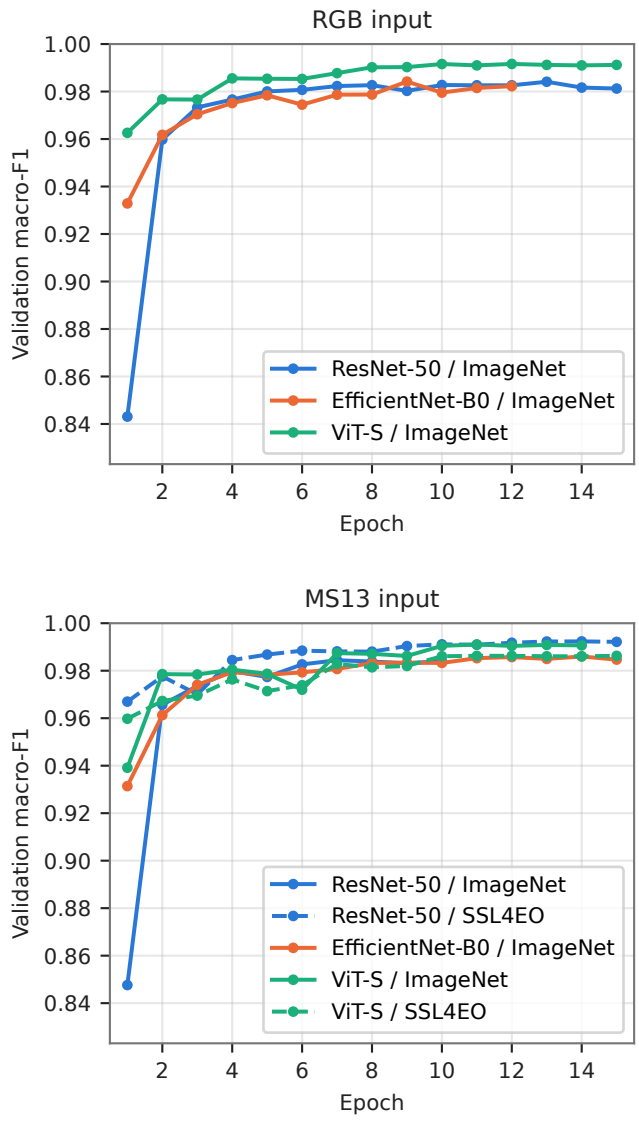


Fig. 2. Validation macro-F1 on EuroSAT per training epoch. Top: RGB input. Bottom: 13-band (MS13) input; dashed lines are the SSL4EO-S12 runs.

open cover crossed by tracks. April falls in the dry season [1], so rainfed fields are bare or fallow, managed grass is brown, and natural cover is xeric scrub on dry, rocky ground, which resembles neither European forest nor European grassland. This is consistent with Fig. 3(a), where Forest and Pasture, two classes whose European look depends on green vegetation, were at or near zero in every run, and with Fig. 4(a), where Forest patches went to HerbaceousVegetation. AnnualCrop, whose EuroSAT patches already include pale, bare parcels (Fig. 1(b)), kept a nonzero F1 in every run. Some objects also have no EuroSAT counterpart. The solar park in Fig. 6(b), which we labeled Industrial, looks nothing like the warehouse roofs of EuroSAT, and a model can only answer with one of the classes it knows.

B. Why fine-tuning helped only in part (RQ-c)

A likely reason for the small gain of `head` lies in the features. If the frozen EuroSAT features had separated the Aguascalientes classes, a new linear classifier trained on 561 labeled patches would likely have found the boundaries. It did not, which suggests that the differences that matter here, such as canopy density on dry ground or a faint parcel pattern on bare soil, were not well represented in features learned to separate European classes. A linear layer can redraw boundaries, but it cannot create a feature the backbone does not compute.

A second explanation is that the classifier was undertrained. With 561 patches, a batch of 64 and 10 epochs, `head` took only 90 optimizer steps, starting at a learning rate of 10^{-4} that then decayed with the cosine schedule. The training logs agree: in 7 of the 8 runs, the `head` training loss was lower at the last epoch than at the middle one, so it had not yet leveled off. In other words, a longer `head` training might have gained more. Both explanations predict a small gain, so our runs cannot separate them; a `head` run trained until its loss stops falling would.

`last_block` adapts the high-level features themselves, which is consistent with its larger gains. Still, the largest part of the drop remained, and we see three likely limits. First, 561 labels are few, and several classes have a few dozen fine-tune patches or fewer (Table I). Second, the 10 fixed epochs had no validation split to choose a better stopping point. Third, in the two CNNs the frozen layers kept their EuroSAT BatchNorm statistics, so the spectral shift may still have passed through them before reaching the trained block. The mode also has a confound: it trains 64 %, 28 % and 8 % of the parameters of ResNet-50, EfficientNet-B0 and ViT-S, so the recovery of different architectures is not a like-for-like comparison.

The ResNet-50/RGB/ImageNet run, where `last_block` gave a lower macro-F1 than `head`, has two possible explanations. It trained the largest share, 64 % of a model of the size listed in Table VI, on 561 patches, which leaves room to fit the fine-tune split instead of the region. The class-weighted loss of Eq. (1) also pushed predictions away from HerbaceousVegetation, whose F1 fell more in this run than in any other (Fig. 3(b)). However, the two 13-band ResNet-50 runs trained the same share with the same loss and did gain from `last_block`, so neither reason explains the run alone. With one seed, we cannot separate them from an unlucky run.

C. CNNs versus the Vision Transformer (RQ-a)

On EuroSAT the architecture made little difference (Table III), probably because the benchmark is close to its ceiling for all three models. In Aguascalientes, among the RGB ImageNet runs, ViT-S had the highest observed macro-F1 in the zero-shot, `head` and `last_block` columns of Table IV. The first two columns are free of the `last_block` confound, because there the three models train the same part: nothing, or only the classifier. With 13 bands and ImageNet weights, however, ResNet-50 had the highest observed value

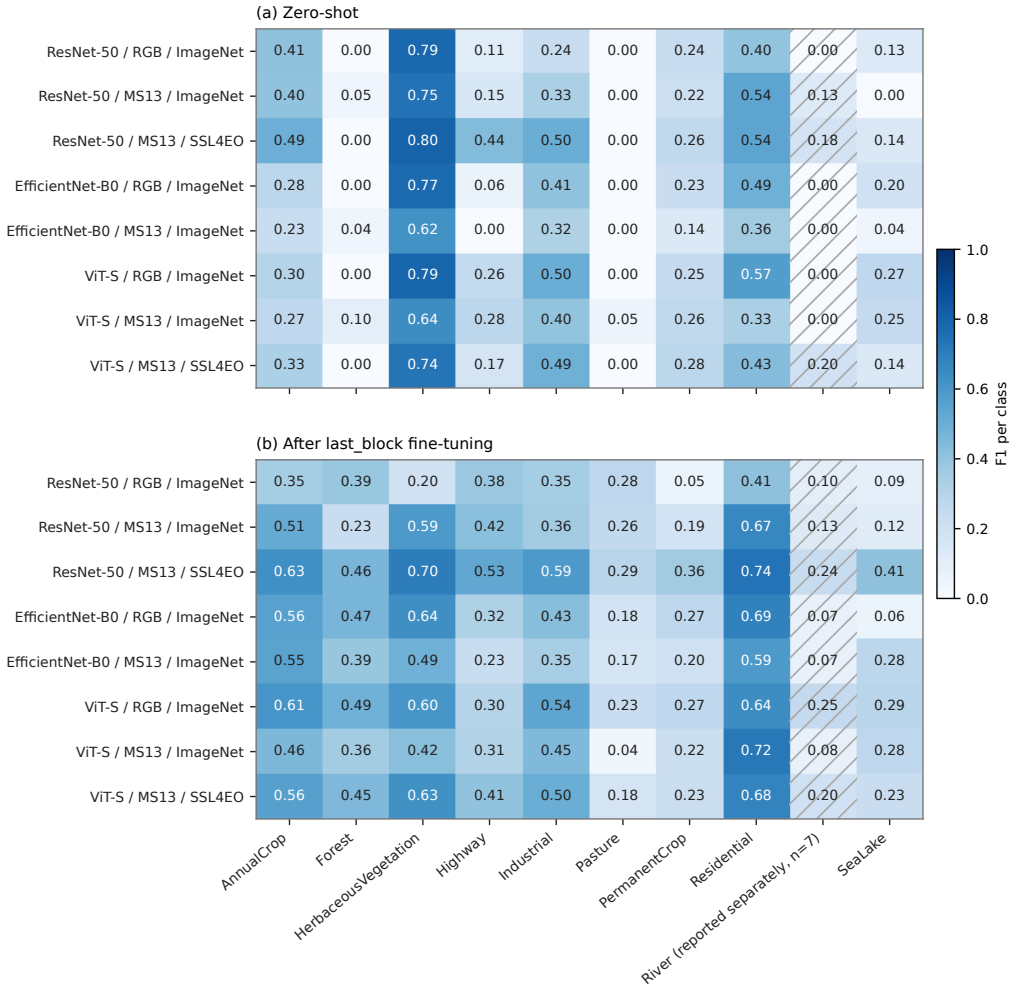


Fig. 3. Per-class F1 on the Aguascalientes test split for the eight runs. (a) Zero-shot. (b) After `last_block` fine-tuning. River (hatched) has too few test patches and is reported separately.

after `last_block`, by a small margin over the other two, so the ViT-S lead is tied to the RGB input in our data.

One hypothesis is global context against local texture. In a ViT, attention derives long-range relations between pixels [7], while a CNN builds its features from local neighborhoods. If the arrangement of a scene, such as parcel boundaries or a street grid, carried over to Aguascalientes better than its color and texture, a model that weighs the whole patch would lose less. The Grad-CAM maps are consistent with this: ViT-S maps spread over the whole patch, while CNN maps cover one compact region. We cannot confirm the hypothesis, for two reasons besides the single seed (Section VI-G). Grad-CAM on a ViT, computed on `norm1` of the last block, is an approximation, so part of the spread may come from the method. And `last_block` trains only 8 % of ViT-S. With 561 labels, training fewer parameters may help or hurt, so this confound could favor either family.

D. RGB versus 13 bands and the pre-trained weights (RQ-d, RQ-e)

With ImageNet weights, the non-RGB bands entered through filters initialized as the mean of the RGB filters, so they had no pre-trained meaning. Whatever they learned came from EuroSAT alone, and the SWIR bands, among the most shifted in our screening, carried that shift straight into the network. This is consistent with EfficientNet-B0 and ViT-S scoring lower with 13 bands. ResNet-50 is the exception, and our runs cannot tell why. The answer to RQ-d therefore depends on the model.

The SSL4EO-S12 weights were learned on all 13 bands, from images sampled worldwide around populated cities in four seasons [11]. Every band thus had a pre-trained meaning, and the pre-training may have included dry landscapes and spectra closer to Aguascalientes than EuroSAT. This is consistent with the higher macro-F1 of both 13-band SSL4EO-S12 runs after `last_block`. A possible reason why the ViT-S gain appeared only after `last_block`, and not

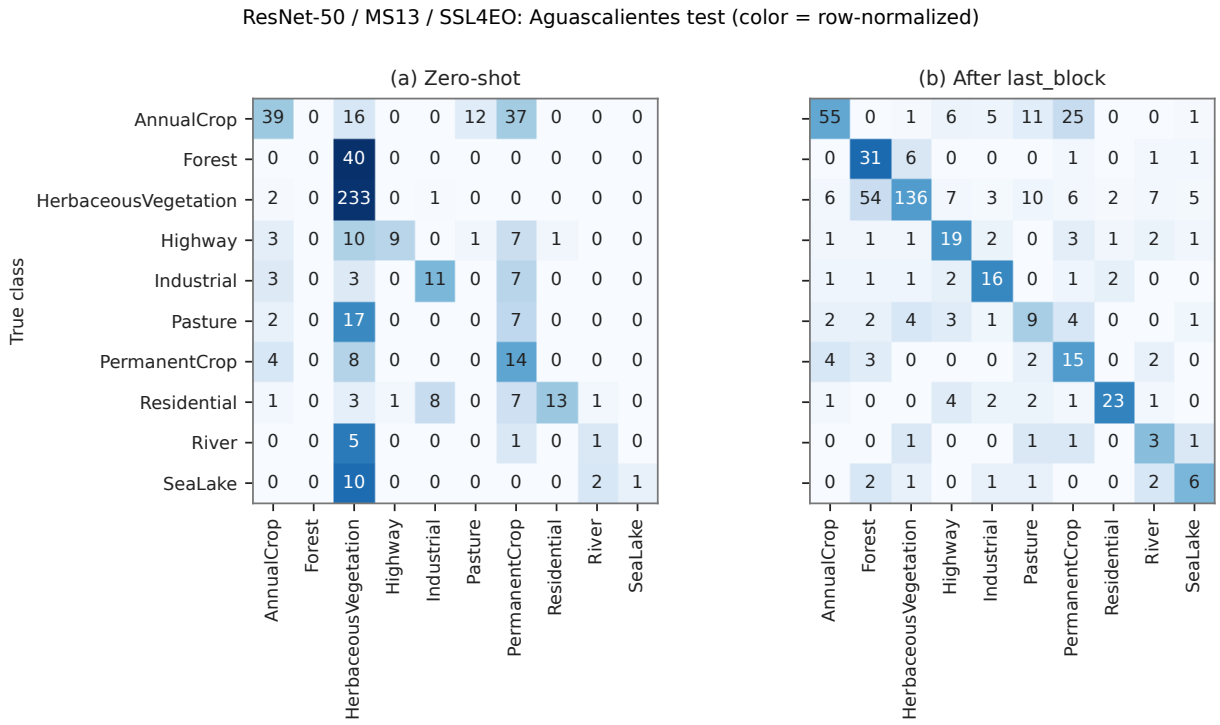


Fig. 4. Confusion matrices of the selected model, ResNet-50/MS13/SSL4EO, on the Aguascalientes test split. (a) Zero-shot. (b) After `last_block` fine-tuning. Cells show patch counts; the color is normalized per row (true class).

zero-shot, is that EuroSAT training mapped the features to European classes, and only local training re-mapped them. Two facts keep us from crediting the weights. The SSL4EO-S12 runs also used another normalization, $x/10{,}000$, which changes how the shifted bands reach the first layer. And ViT-S/MS13/SSL4EO was slightly lower than its ImageNet counterpart on EuroSAT, so the benefit is not uniform.

E. What the Grad-CAM maps suggest

The placement of the Aguascalientes maps described in Section V is consistent with models looking for European cues, such as green vegetation, that these patches lack. The solar park of Fig. 6(b) is an out-of-distribution object: no model centered its map on the panels, and none predicted Industrial. Fig. 7 adds one encouraging case: after `last_block`, the selected model predicted the AnnualCrop patch correctly, and its map moved from one corner to a wider part of the fields. One patch, however, shows a possible behavior, not a general one.

The main error after fine-tuning, HerbaceousVegetation predicted as Forest (Fig. 4(b)), has a likely reason in the landscape. The two such patches in Fig. 7 show rough ground with dark areas, and one is almost entirely dark. In Aguascalientes, Forest on the western ranges stays dark even in April, so xeric scrub on rugged hills shares its two most visible cues: dark tone and rough relief. The guide separates them with a threshold, continuous canopy over 70 % of the cell, on what is in reality a gradual change from scrub to open woodland to forest. The class weights may add to it: Forest has far fewer fine-tune patches than HerbaceousVegetation (Table I), so it gets a much larger weight in Eq. (1), and borderline patches are pushed toward it.

F. Cost (RQ-a)

Table VI shows that EfficientNet-B0 has far fewer parameters than the other two models but a similar time per epoch. We see two possible reasons. First, on the MPS backend the time may depend more on the number of layers and on moving activations through memory than on arithmetic, and EfficientNet-B0 has many small layers. Second, the three models may share a bottleneck in data loading. Every run used the same loader with two worker processes, so if the loader could not feed the GPU fast enough, all three models would wait about the same time for data at each step, whatever their size. We did not profile the runs or time the loader on its own, so we cannot tell the two reasons apart. For the second stage of the project, monthly inference over the whole state, the useful number is patches/s; there, ViT-S/RGB/ImageNet had the highest observed throughput, and the 13-band input cost throughput in every model.

G. Limitations

Our results have eight limits. First, each configuration was trained once with seed 0 and no paired test was run, so small gaps between runs may change with another seed. Second, we used one month, April 2024, so season and region are mixed in the shift and the rainy season is not covered. Third, a single annotator labeled the cells, with no agreement measure; the

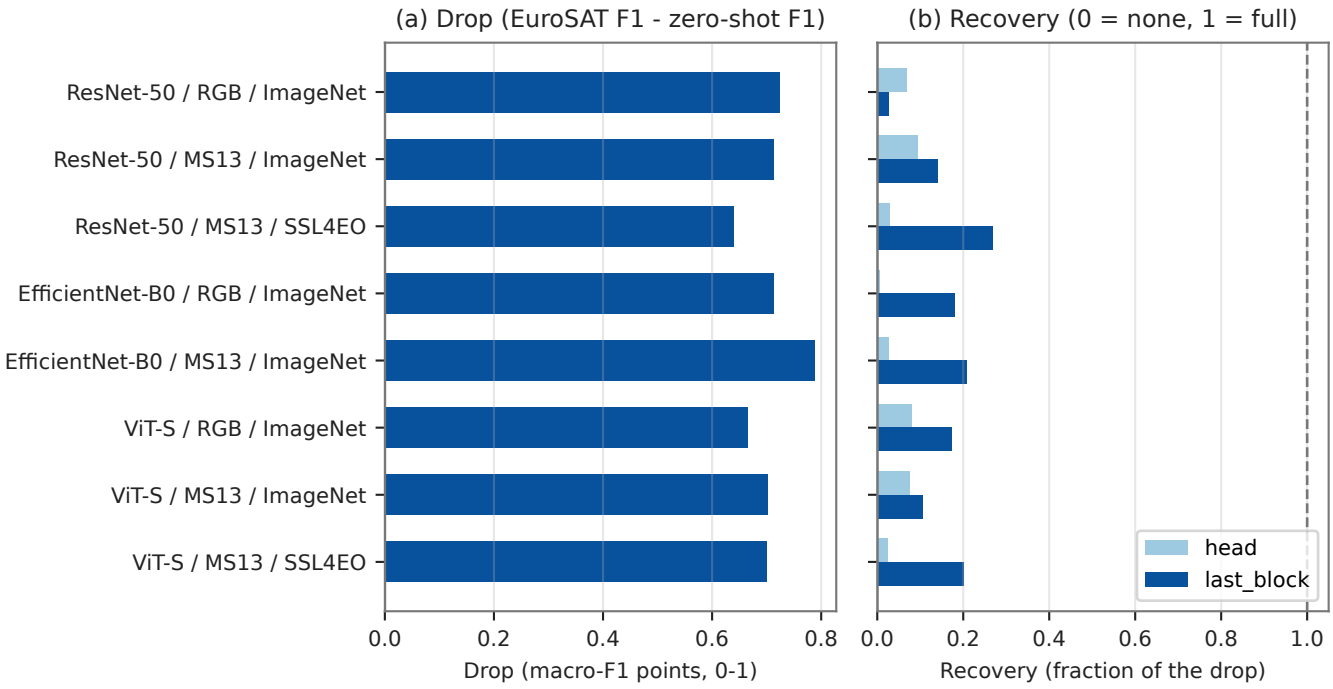


Fig. 5. (a) Drop from EuroSAT macro-F1 to Aguascalientes zero-shot macro-F1 for each run. (b) Recovery after `head` and `last_block` fine-tuning, as a fraction of the drop; the dashed line marks full recovery.

boundaries between scrub, open woodland and pasture depend on judgment. Fourth, the test set has 541 patches and the fine-tune set 561, River has only 7 test patches, and several classes a few dozen or fewer, so per-class F1 is noisy; skipped covers, such as greenhouses and quarries, are not measured at all. Fifth, 35 test patches border a fine-tune cell because the split has no buffer zone, so the fine-tuned scores may be slightly optimistic. Sixth, the 64-px patches were resized to 224 pixels, which adds no detail and no context beyond the cell. Seventh, we used L1C data without atmospheric correction, so part of the spectral shift, as B10 suggests, may be atmospheric. Eighth, EfficientNet-B0 had no SSL4EO-S12 weights, so RQ-e was answered for two models only.

H. Future work

The most direct next step is to train each configuration with several seeds, which would allow a paired test between models. A buffer zone between fine-tune and test blocks would remove the adjacency risk, and a fine-tuning mode that trains the same share of parameters in every model would remove the main confound of the CNN versus ViT comparison. Multi-season composites, which are part of the planned second stage of the project, would separate the effect of the dry season from that of the region. More local labels, from a second annotator, would both help fine-tuning and measure label noise. Finally, other self-supervised Sentinel-2 weights and multimodal input, such as radar, are worth testing. Radar sees through clouds and measures structure and moisture, information that the optical bands do not provide.

VII. Conclusions

Our goal was to measure how much a land-use classifier trained on EuroSAT loses in semi-arid Aguascalientes, and how much of that loss light fine-tuning on local labels wins back. We compared ResNet-50, EfficientNet-B0 and ViT-S/16 under one shared protocol, and three findings stand out.

1) *The models were almost tied on EuroSAT, but all of them dropped sharply in Aguascalientes (RQ-a, RQ-b).* EuroSAT macro-F1 ranged from 0.983 to 0.993, and without local training the drop ranged from 0.639 to 0.788. This means that a near-perfect EuroSAT score does not guarantee a useful model in another region; the model has to be evaluated on local labels.
2) *Light fine-tuning won back only part of the drop (RQ-c).* Training the classifier alone (`head`) recovered at most 0.095 of the drop, and our runs cannot tell whether the features were too weak or the classifier was undertrained. Fine-tuning the last block (`last_block`) recovered between 0.026 and 0.268 (Eq. 3). This means that most of the drop remained in every run after fine-tuning.
3) *The effect of the input bands and the weights depended on the model (RQ-d, RQ-e).* With ImageNet weights, 13 bands helped only ResNet-50 in Aguascalientes. With 13 bands, the SSL4EO-S12 runs reached a higher macro-F1 after `last_block` than the ImageNet runs, for both ResNet-50 and ViT-S, but they also used another normalization, so the gap cannot be credited to the

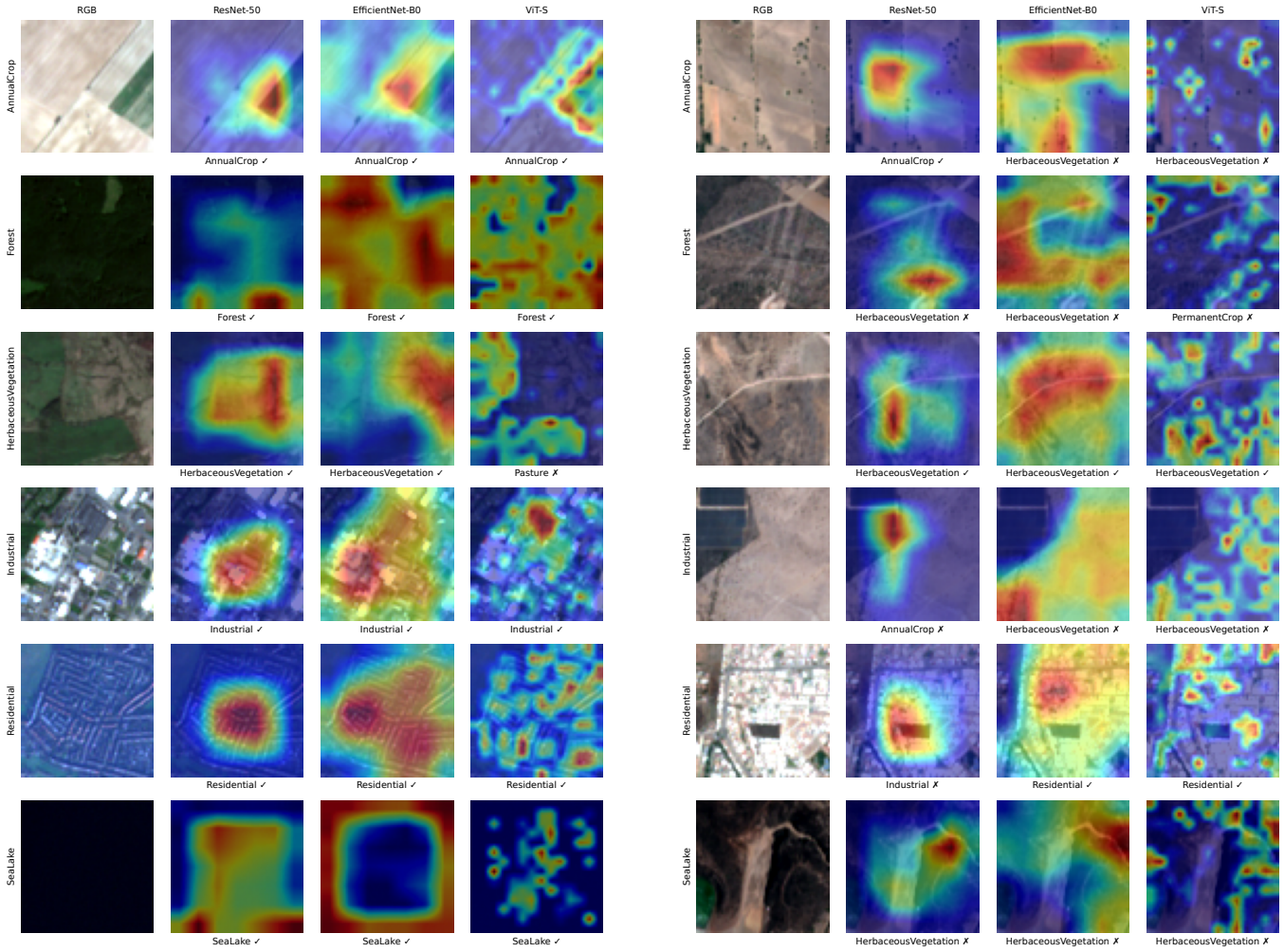


Fig. 6. Grad-CAM maps of the RGB ImageNet runs, without fine-tuning, on one random test patch per class (seed 0). (a) Left: EuroSAT. (b) Right: Aguascalientes. Columns: RGB patch, ResNet-50, EfficientNet-B0 and ViT-S. Under each map: the predicted class, with a check mark if correct and a cross if wrong.

weights alone. This means that the answer depends on the combination of model, bands and weights, not only on CNN versus ViT.

The selection rule, fixed before any result, chose ResNet-50/MS13/SSL4EO with `last_block` fine-tuning, with a macro-F1 of 0.525, 0.084 above the runner-up. Among the six ImageNet runs, where the three models compete on equal weights, ViT-S/RGB/ImageNet came first with 0.441, 0.040 above the next run. On cost (RQ-a), EfficientNet-B0 had far fewer parameters but a similar time per epoch, and ViT-S/RGB/ImageNet had the highest observed throughput, 236 patches/s.

On the method side, we learned that sharing every hyperparameter did not make the fine-tuning comparison like-for-like, because the same `last_block` mode trained a different share of each model; a fair comparison has to match what is trained, not only how. Next, the selected model will be the starting point for the second stage of the project, a monthly land-use monitor of the state, and the experiments in Section VI, starting with several seeds per configuration, would make these answers firmer.

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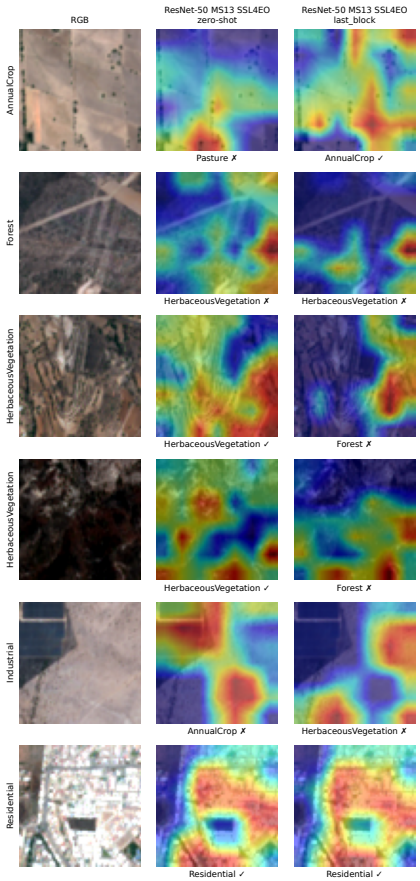


Fig. 7. Grad-CAM maps of the selected model, ResNet-50/MS13/SSL4EO, on Aguascalientes test patches. Columns: RGB patch, zero-shot and after `last_block`. The two HerbaceousVegetation rows are random patches (seed 0) among those predicted as Forest after `last_block`; the other rows are the patches of Fig. 6(b).

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